

Demand Model Specification

EMS Optimization Project - Phase 2: Demand Modeling

Last Updated: March 12, 2026

Executive Summary

This document specifies the stochastic demand models fitted for crash arrivals in the EMS discrete-event simulation. Based on 416,434 Manhattan crash records (2012-2026), we developed:

- 1. **Baseline Homogeneous Poisson** - Constant rate model (rejected by chi-square test)
- 2. **Non-Homogeneous Poisson Process (NHPP)** - Recommended model with hourly and day-of-week variation
- 3. **Spatial Demand Model** - Precinct-level arrival rates

1. Data Summary

Metric	Value
Total crashes	416,434
Date range	July 1, 2012 - February 20, 2026
Duration	4,983 days (119,592 hours)
Average daily rate	83.6 crashes/day
Average hourly rate	3.48 crashes/hour

Geographic Distribution

- **CBD crashes:** 231,786 (55.7%)
- **Non-CBD crashes:** 184,648 (44.3%)

2. Homogeneous Poisson Model

Model Specification

$$N(t) \sim \text{Poisson}(\lambda t)$$

Where:

- $\lambda = 3.482$ crashes/hour (constant)

Goodness-of-Fit Test

Test	Statistic	p-value	Result
Chi-square	723,713.09	< 0.0001	REJECT
Dispersion (Var/ Mean)	2.43	-	Overdispersed

Conclusion: The homogeneous Poisson assumption is **rejected**. The data shows significant time-varying patterns that require a non-homogeneous model.

Limitations

- Ignores clear hourly and daily patterns
- Underestimates demand during peak hours
- Overestimates demand during off-peak hours
- Not suitable for realistic simulation

3. Non-Homogeneous Poisson Process (NHPP)

Model Specification

The recommended model uses a time-varying arrival rate:

$$\lambda(t) = \lambda_{\text{base}} \times f_{\text{hour}}(h) \times f_{\text{dow}}(d)$$

Where:

- λ_{base} = 3.482 crashes/hour (overall mean rate)
- $f_{\text{hour}}(h)$ = hourly factor for hour $h \in \{0, 1, \dots, 23\}$
- $f_{\text{dow}}(d)$ = day-of-week factor for day $d \in \{0=\text{Mon}, 1=\text{Tue}, \dots, 6=\text{Sun}\}$

3.1 Hourly Factors

Hour	λ (crashes/hr)	Factor	Description
0	3.46	0.88	Midnight
1	2.42	0.61	
2	2.08	0.53	
3	1.93	0.49	
4	1.94	0.49	
5	1.78	0.45	Minimum
6	2.17	0.55	
7	2.61	0.66	Morning ramp-up
8	4.24	1.08	Morning commute
9	4.81	1.22	
10	4.83	1.23	
11	5.03	1.28	
12	5.15	1.31	Noon
13	5.27	1.34	
14	5.76	1.47	Afternoon
15	5.20	1.32	
16	5.97	1.52	Peak
17	5.70	1.45	Evening commute
18	5.28	1.34	
19	4.60	1.17	
20	4.04	1.03	
21	3.60	0.91	
22	3.44	0.87	
23	3.10	0.79	

Peak Period: 2 PM - 6 PM (factors > 1.3)

Low Period: 3 AM - 6 AM (factors < 0.5)

3.2 Day-of-Week Factors

Day	λ (crashes/day)	Factor
Monday	78.2	0.94
Tuesday	86.4	1.03
Wednesday	87.1	1.04
Thursday	90.3	1.08
Friday	95.8	1.15
Saturday	79.4	0.95
Sunday	67.9	0.81

Key Insight: Friday has 41% more crashes than Sunday.

3.3 Combined Rate Calculation

Example: Friday at 5 PM

$$\lambda(\text{Friday}, 17:00) = 3.482 \times 1.45 \times 1.15 = 5.81 \text{ crashes/hour}$$

Example: Sunday at 5 AM

$$\lambda(\text{Sunday}, 05:00) = 3.482 \times 0.45 \times 0.81 = 1.27 \text{ crashes/hour}$$

4. Spatial Demand Model

Precinct-Level Rates

Precinct	Crashes	λ/hour	λ/day	% Total	Category
19 (Midtown East)	41,097	0.344	8.25	9.9%	High
18 (Midtown North)	35,577	0.297	7.14	8.5%	High
14 (Midtown South)	31,551	0.264	6.33	7.6%	High
1 (Financial District)	27,999	0.234	5.62	6.7%	High
17 (Midtown East)	27,628	0.231	5.54	6.6%	High
13 (Lower East Side)	25,101	0.210	5.04	6.0%	High
10 (Chelsea)	23,796	0.199	4.78	5.7%	High
...	...	...	...	...	...
22 (Central Park)	4,209	0.035	0.84	1.0%	Low
50 (Marble Hill)	765	0.006	0.15	0.2%	Low
52 (Inwood)	284	0.002	0.06	0.1%	Low
114 (Randall's Island)	80	0.001	0.02	0.0%	Low

High-Demand vs Low-Demand Zones

Category	Precincts	Combined λ/day
High ($\geq 75\text{th percentile}$)	19, 18, 14, 1, 17, 13, 10	46.7 (55.9%)
Low ($\leq 25\text{th percentile}$)	28, 30, 26, 22, 50, 52, 114	5.0 (6.0%)

5. CBD vs Non-CBD Patterns

Overall Rates

Area	λ (crashes/hour)	λ (crashes/day)	% of Total
CBD	1.94	46.5	55.7%
Non-CBD	1.54	37.1	44.3%

Temporal Pattern Differences

Characteristic	CBD	Non-CBD
Peak hour	14:00 (2 PM)	16:00 (4 PM)
Peak factor	1.60	1.71
Minimum hour	4 AM	3 AM
Peak day	Friday	Friday
Weekend dropoff	Sharper	Moderate

Insight: CBD peaks earlier (business hours) while non-CBD peaks during evening commute.

6. Implementation for Simulation

Arrival Generation Algorithm

```

import numpy as np
import pandas as pd

# Load rate tables
hourly_factors = pd.read_csv('demand_lambda_hourly.csv').set_index('hour')['factor']
dow_factors = pd.read_csv('demand_lambda_dow.csv').set_index('dow')['factor']
precinct_probs = pd.read_csv('demand_lambda_precinct.csv').set_index('precinct')['pct_of_total'] / 100

BASE_RATE = 3.482 # crashes per hour

def get_arrival_rate(hour: int, dow: int, area: str = 'all') -> float:
    """
    Get arrival rate  $\lambda(t)$  for given time.

    Args:
        hour: Hour of day (0-23)
        dow: Day of week (0=Monday, 6=Sunday)
        area: 'all', 'cbd', or 'non_cbd'

    Returns:
        Arrival rate in crashes per hour
    """
    rate = BASE_RATE * hourly_factors[hour] * dow_factors[dow]

    if area == 'cbd':
        rate *= 0.557 # CBD proportion
    elif area == 'non_cbd':
        rate *= 0.443 # Non-CBD proportion

    return rate

def generate_next_arrival(current_time: float, dt: float = 1.0) -> float:
    """
    Generate next arrival time using thinning algorithm.

    Args:
        current_time: Current simulation time (hours from start)
        dt: Time step for rate evaluation

    Returns:
        Time until next arrival (hours)
    """
    # Get maximum rate in next period
    hour = int(current_time % 24)
    dow = int((current_time // 24) % 7)
    lambda_max = get_arrival_rate(hour, dow) * 1.1 # Safety margin

    t = 0
    while True:
        # Generate candidate arrival
        t += np.random.exponential(1 / lambda_max)

        # Accept/reject
        new_hour = int((current_time + t) % 24)
        new_dow = int(((current_time + t) // 24) % 7)
        lambda_t = get_arrival_rate(new_hour, new_dow)

        if np.random.random() < lambda_t / lambda_max:
            return t

def assign_precinct() -> int:

```



```
"""Randomly assign crash to precinct based on demand distribution."""
return np.random.choice(precinct_probs.index, p=precinct_probs.values)
```

Validation Checks

Before simulation, verify:

1. Generated hourly counts match expected patterns (within 10%)
2. Day-of-week distribution matches historical (within 5%)
3. Precinct assignments match spatial distribution

7. Model Limitations

1. **Stationarity assumption:** Factors assumed constant across years
2. **Independence:** Events assumed independent (no clustering)
3. **Weather effects:** Not modeled (could affect rates $\pm 20\%$)
4. **Special events:** Holidays, major events not explicitly modeled
5. **Trend:** Long-term trend not captured (data spans 13+ years)

Recommendations for Future Work

1. Include seasonal (monthly) factors
2. Add weather covariates
3. Model holiday effects
4. Consider Hawkes process for event clustering

8. Seasonal Patterns

Monthly Variation Analysis

Analysis of 416,434 Manhattan crash records reveals moderate seasonal variation in demand:

Month	Factor	Season
January	0.876	Winter
February	0.822	Winter (trough)
March	0.956	Spring
April	0.887	Spring
May	1.067	Spring
June	1.072	Summer
July	1.056	Summer
August	1.049	Summer
September	1.071	Fall
October	1.103	Fall (peak)
November	1.026	Fall
December	1.016	Winter

Statistical Tests

Test	Statistic	p-value	Result
Chi-square (uniformity)	—	< 0.001	Reject
ANOVA (across months)	—	< 0.001	Significant
Kruskal-Wallis	—	< 0.001	Significant

Key Metrics

- **Coefficient of variation:** 9% (moderate)
- **Seasonal amplitude:** 28% (peak-to-trough range)
- **Peak month:** October (factor = 1.103)
- **Trough month:** February (factor = 0.822)

Implications for Simulation

While seasonal variation is statistically significant, its magnitude is **moderate** relative to within-day patterns:

- **Hourly variation:** Factor range 0.5–1.6 (amplitude ~110%)

- **Day-of-week variation:** Factor range 0.85–1.15 (amplitude ~30%)
- **Monthly/seasonal variation:** Factor range 0.82–1.10 (amplitude ~28%)

The NHPP model uses an annual average rate, which is a **reasonable approximation** because:

1. Hourly patterns dominate (10× larger amplitude)
2. The maximum seasonal factor (1.103) is modest
3. Policy rankings are robust to demand multipliers up to 2.0× (Experiment 3), far exceeding seasonal variation

For **high-fidelity future models**, monthly factors could be incorporated as an additional modulation layer in the NHPP intensity function:

$$\lambda(t) = \lambda_{\text{base}} \times f_{\text{hour}}(h) \times f_{\text{dow}}(d) \times f_{\text{month}}(m)$$

Seasonal Visualizations

See `results/figures/seasonal_patterns.png`, `seasonal_decomposition.png`, and `seasonal_heatmap.png` for detailed seasonal analysis plots.

9. Output Files

File	Location	Description
<code>demand_lambda_hourly.csv</code>	<code>data/processed/</code>	24 hourly rates with CBD/non-CBD factors
<code>demand_lambda_dow.csv</code>	<code>data/processed/</code>	7 day-of-week factors
<code>demand_lambda_precinct.csv</code>	<code>data/processed/</code>	Precinct-level rates
<code>demand_model_summary.json</code>	<code>data/processed/</code>	Model summary statistics
<code>fig_demand_model_fit.png</code>	<code>results/figures/</code>	Diagnostic plots
<code>fig_hourly_rates.png</code>	<code>results/figures/</code>	CBD vs Non-CBD comparison
<code>fig_precinct_demand.png</code>	<code>results/figures/</code>	Precinct demand bar chart

References

1. Law, A.M. (2015). Simulation Modeling and Analysis, 5th ed. McGraw-Hill.
2. Ross, S.M. (2014). Introduction to Probability Models, 11th ed. Academic Press.
3. NYC Open Data - Motor Vehicle Collisions (NYPD)